

Advanced (Habanero)

- Q1.** What is the vertex of the graph of $y = |x - 2|$?
(A) $(2, 0)$
(B) $(0, 2)$
(C) $(2, 2)$
(D) $(0, 0)$
(E) $(1, 1)$
- Q2.** What is the axis of symmetry of the graph of $y = |x + 1|$?
(A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $y = -1$
(E) $y = 0$
- Q3.** Which of the following is the graph of $y = 2|x|$?
(A) A graph with a vertex at $(0, 0)$ and opening upwards
(B) A graph with a vertex at $(0, 0)$ and opening downwards
(C) A graph with a vertex at $(0, 2)$ and opening upwards
(D) A graph with a vertex at $(0, 2)$ and opening downwards
(E) A graph with a vertex at $(2, 0)$ and opening upwards
- Q4.** What is the equation of the graph of $y = |x - 3| + 2$?
(A) $y = |x - 3| - 2$
(B) $y = |x + 3| + 2$
(C) $y = |x - 3| + 2$
(D) $y = |x + 3| - 2$
(E) $y = |x - 2| + 3$
- Q5.** Which of the following graphs represents the equation $y = -|x + 2|$?
(A) A graph with a vertex at $(-2, 0)$ and opening downwards
(B) A graph with a vertex at $(-2, 0)$ and opening upwards
(C) A graph with a vertex at $(2, 0)$ and opening downwards
(D) A graph with a vertex at $(2, 0)$ and opening upwards
(E) A graph with a vertex at $(0, 0)$ and opening downwards
- Q6.** What is the equation of the graph of $y = 3|x - 2|$?
(A) $y = 3|x + 2|$
(B) $y = 3|x - 2|$
(C) $y = |3x - 2|$
(D) $y = |3x + 2|$
(E) $y = 2|x - 3|$
- Q7.** Which of the following is the graph of $y = |x| - 2$?
(A) A graph with a vertex at $(0, -2)$ and opening upwards
(B) A graph with a vertex at $(0, 0)$ and opening downwards
(C) A graph with a vertex at $(0, 2)$ and opening upwards
(D) A graph with a vertex at $(0, 2)$ and opening downwards
(E) A graph with a vertex at $(2, 0)$ and opening upwards
- Q8.** What is the axis of symmetry of the graph of $y = |2x - 3|$?
(A) $x =$
- Q9.** Which of the following graphs represents the equation $y = 0$?
(A) $x =$
(B) $y = -1$

Open Ended Questions (FRQ)

- Q9.** Graph the equation $y = |x + 1| - 3$. Identify the vertex, axis of symmetry, and direction of opening.
- Q10.** Write the equation of the absolute value function with a vertex at $(2, 1)$ and an axis of symmetry at $x = 2$. Then graph the function.